EE-102: Electric Circuit Analysis

LECTURE NO 5:
Methods of Analysis

Engr. Tehseen Ahsan
Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan

Organization of Lecture

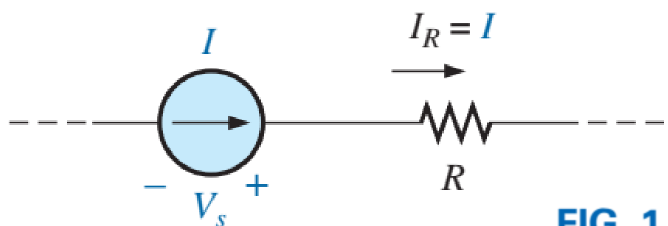
1. Introduction
2. Current Sources
3. Source Conversions
4. Current Sources in Parallel
5. Current Sources in Series
6. Branch-Current Analysis
7. Mesh Analysis
8. Nodal Analysis
9. Bridge Networks
10. Y-Δ and Δ-Y Conversions

1. Introduction

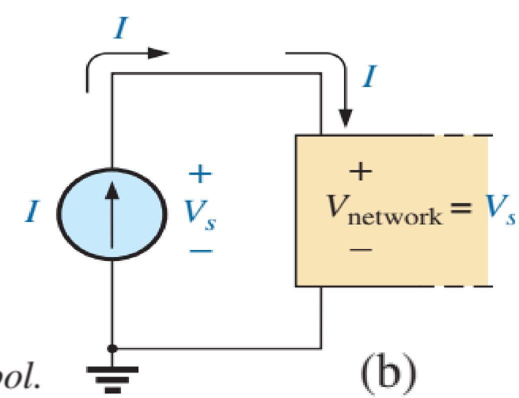
- The circuits you previously studied had only one source or two or more sources in series or parallel. The step-by-step procedures can be applied only if the sources are in series or parallel. *There will be an interaction of sources that will not permit the reduction techniques used to find quantities such as the total resistance and the source current*.
- For such situations, methods of analysis have been developed that allow us to approach, in a systematic manner, networks with any number of sources in any arrangement. To our benefit, the methods to be introduced can also be applied to networks with only *one source* or to networks in which sources are in *series or parallel*.
- The methods to be introduced in this chapter include branch-current analysis, mesh analysis, and nodal analysis. Each can be applied to the same network, although usually one is more appropriate than the other. The "best" method cannot be defined by a strict set of rules but can be determined only after developing an understanding of the relative advantages of each.

2. Current Sources (1/4)

- The current source is often described as the ***dual*** of the voltage source. Just as a battery provides a fixed voltage to a network, ***a current source establishes a fixed current in the branch where it is located***.
- The term *dual* is applied to any two elements in which the traits of one variable can be interchanged with the traits of another. This is certainly true for the current and voltage of the two types of sources.
- The symbol for a current source appears in Fig. 1(a). The arrow indicates the direction in which it is supplying current to the branch where it is located. In Fig. 1(b),we find that the voltage across a current source is determined by the polarity of the voltage drop caused by the current source. For single-source networks, it always has the polarity of Fig. 1(b),but for multisource networks it can have either polarity.



(a)



(b)

FIG. 1

Introducing the current source symbol.

2. Current Sources (2/4)

EXAMPLE 1 Find the source voltage, the voltage V_1, and current I_1 for the circuit in Fig. 2.

Solution: Since the current source establishes the current in the branch in which it is located, the current I_1 must equal I, and

$$I_1 = I = \mathbf{10\ mA}$$

The voltage across R_1 is then determined by Ohm's law:

$$V_1 = I_1R_1 = (10\ \text{mA})(20\ \Omega) = \mathbf{200\ V}$$

Since resistor R_1 and the current source are in parallel, the voltage across each must be the same, and

$$V_s = V_1 = \mathbf{200\ V}$$

with the polarity shown.

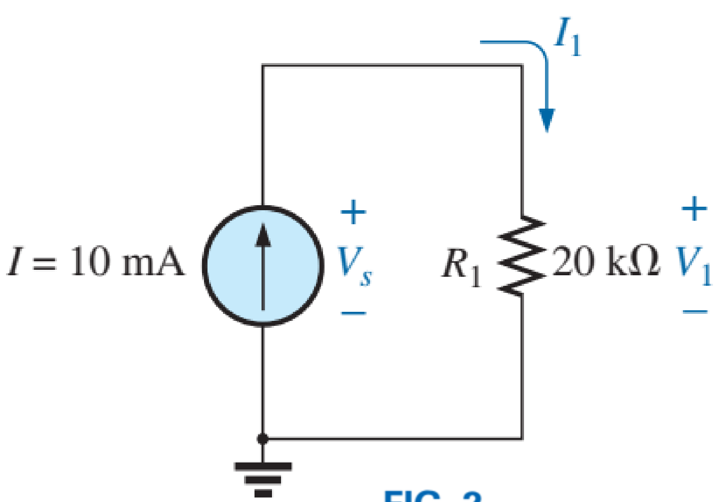


FIG. 2

2. Current Sources (3/4)

EXAMPLE 2 Find the voltage V_s and currents I_1 and I_2 for the network in Fig. 3.

Solution:

Since the current source and voltage source are in parallel,

$$V_s = E = \mathbf{12\ V}$$

Further, since the voltage source and resistor R are in parallel,

$$V_R = E = 12\ \text{V}$$

and

$$I_2 = \frac{V_R}{R} = \frac{12\ \text{V}}{4\ \Omega} = \mathbf{3\ A}$$

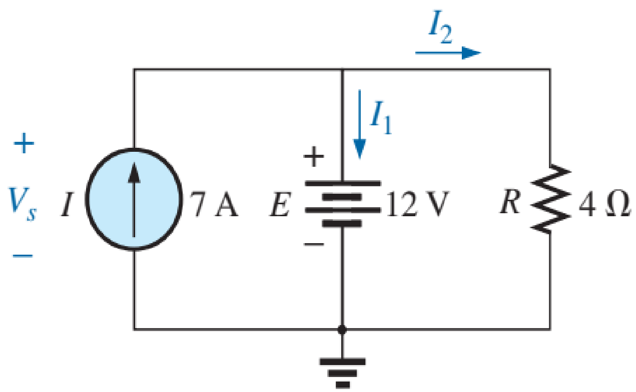


FIG. 3

The current I_1 of the voltage source can then be determined by applying Kirchhoff's current law at the top of the network as follows:

$$\Sigma I_i = \Sigma I_o$$

$$I = I_1 + I_2$$

and

$$I_1 = I - I_2 = 7\ \text{A} - 3\ \text{A} = \mathbf{4\ A}$$

2. Current Sources (4/4)

Homework!

EXAMPLE 3 Determine the current I_1 and the voltage V_s for the network in Fig. 4.

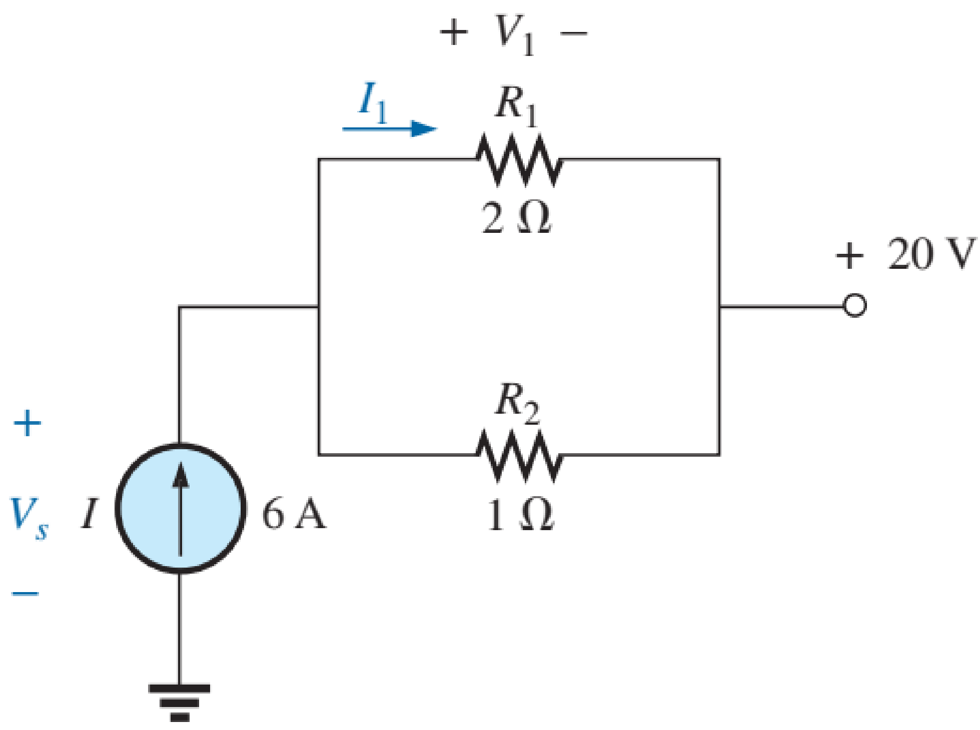


FIG. 4

3. Source Conversions (1/6)

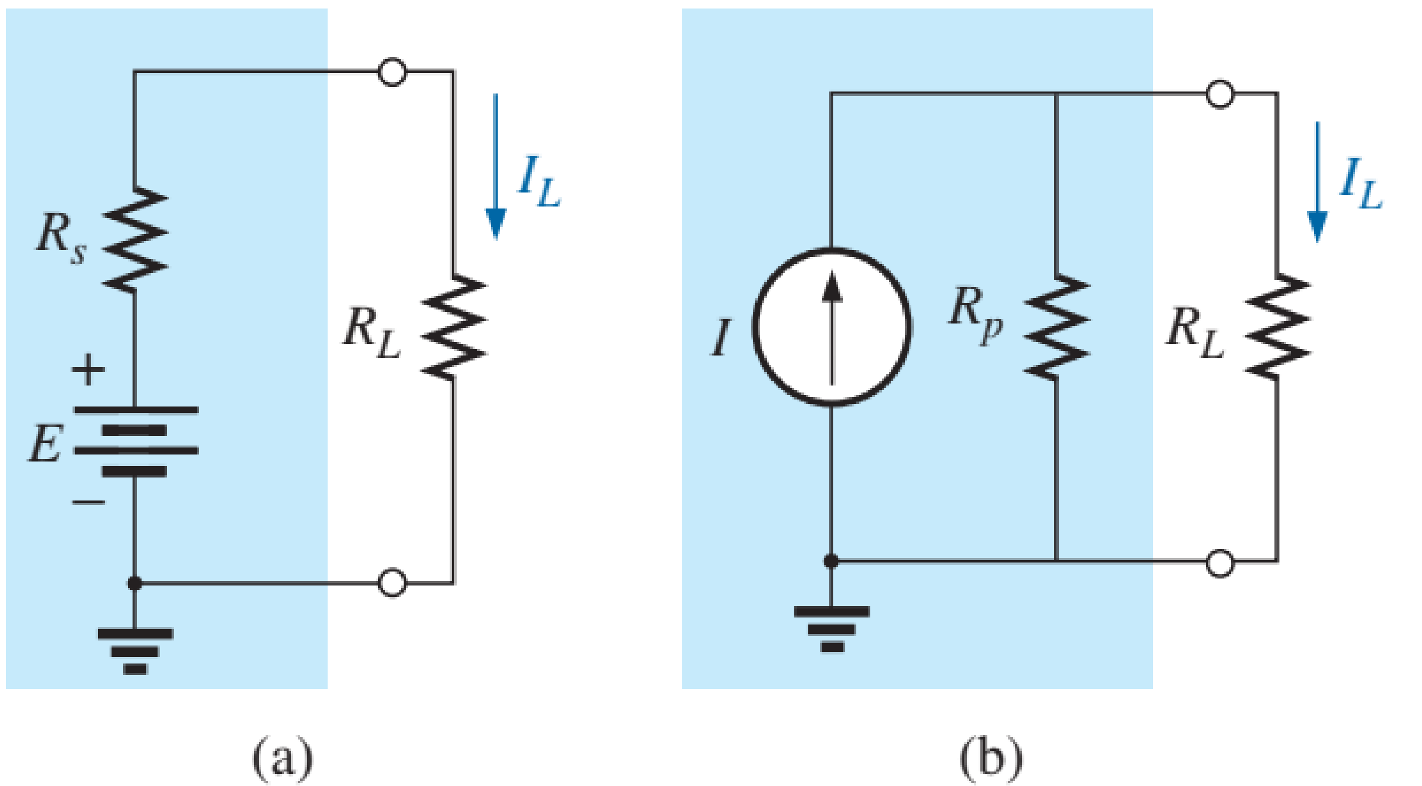


FIG. 5

Practical sources: (a) voltage; (b) current.

3. Source Conversions (2/6)

- Equivalence can be established using the equations appearing in Fig. 6. First note that the resistance is the same in each configuration—a nice advantage. For the voltage source equivalent, the voltage is determined by a simple application of Ohm's law to the current source: $E = IR_p$.
- For the current source equivalent, the current is again determined by applying Ohm's law to the voltage source: $I = E/R_s$ .At first glance, it all seems too simple, but Example 4 next slide verifies the results.

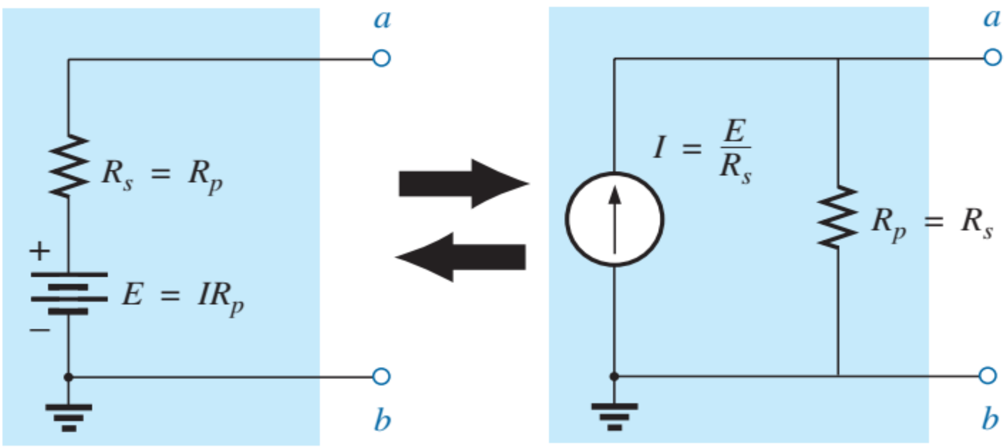


FIG. 6
Source conversion.

3. Source Conversions (3/6)

EXAMPLE 4 For the circuit in Fig. 7:

a. Determine the current I_L.
b. Convert the voltage source to a current source.
c. Using the resulting current source of part (b), calculate the current through the load resistor, and compare your answer to the result of part (a).

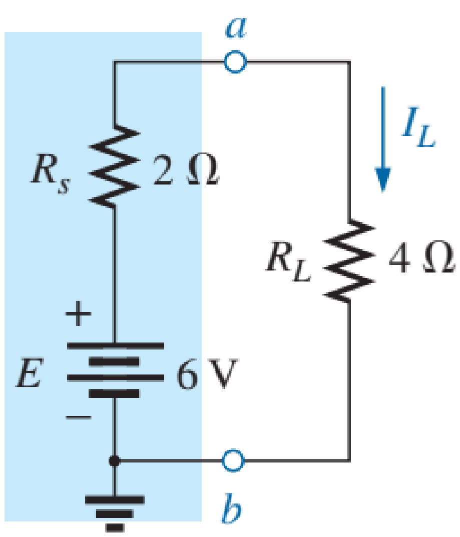


FIG. 7

Practical voltage source and load for Example 4.

3. Source Conversions (4/6)

Solutions:

a. Applying Ohm's law gives

$$I_L = \frac{E}{R_s + R_L} = \frac{6\text{ V}}{2\ \Omega + 4\ \Omega} = \frac{6\text{ V}}{6\ \Omega} = \mathbf{1\ A}$$

b. Using Ohm's law again gives

$$I = \frac{E}{R_s} = \frac{6\text{ V}}{2\ \Omega} = \mathbf{3\ A}$$

and the equivalent source appears in Fig. 8 with the load reapplied.

c. Using the current divider rule gives

$$I_L = \frac{R_p I}{R_p + R_L} = \frac{(2\ \Omega)(3\text{ A})}{2\ \Omega + 4\ \Omega} = \frac{1}{3}(3\text{ A}) = \mathbf{1\ A}$$

We find that the current I_L is the same for the voltage source as it was for the equivalent current source—the sources are therefore equivalent.

3. Source Conversions (5/6)

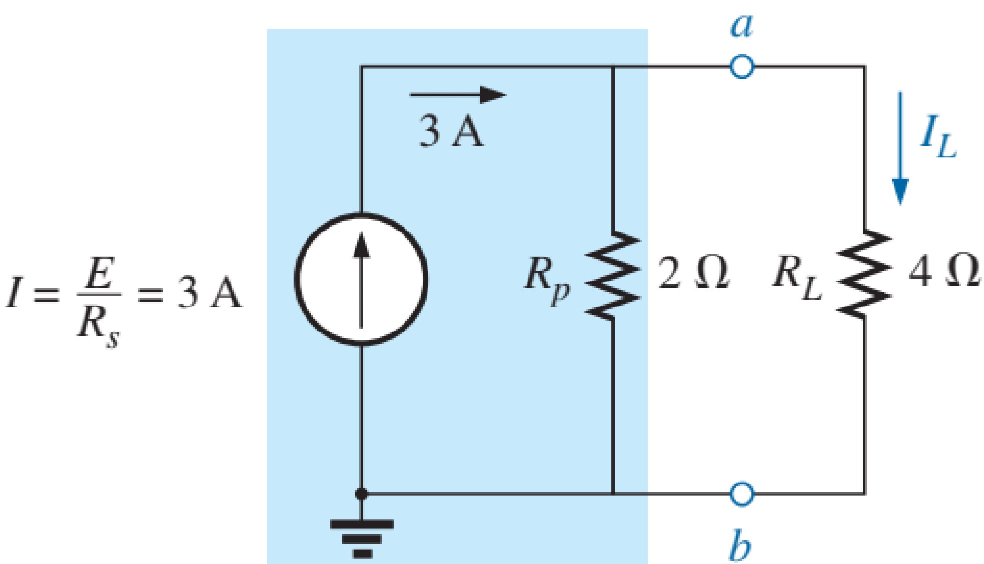


FIG. 8

Equivalent current source and load for the voltage source in Fig. 7.

As demonstrated in Fig. 5 and in Example 4, note that

a source and its equivalent will establish current in the same direction through the applied load.

3. Source Conversions (6/6)

Homework!

EXAMPLE 5 Determine current I_2 for the network in Fig. 9.

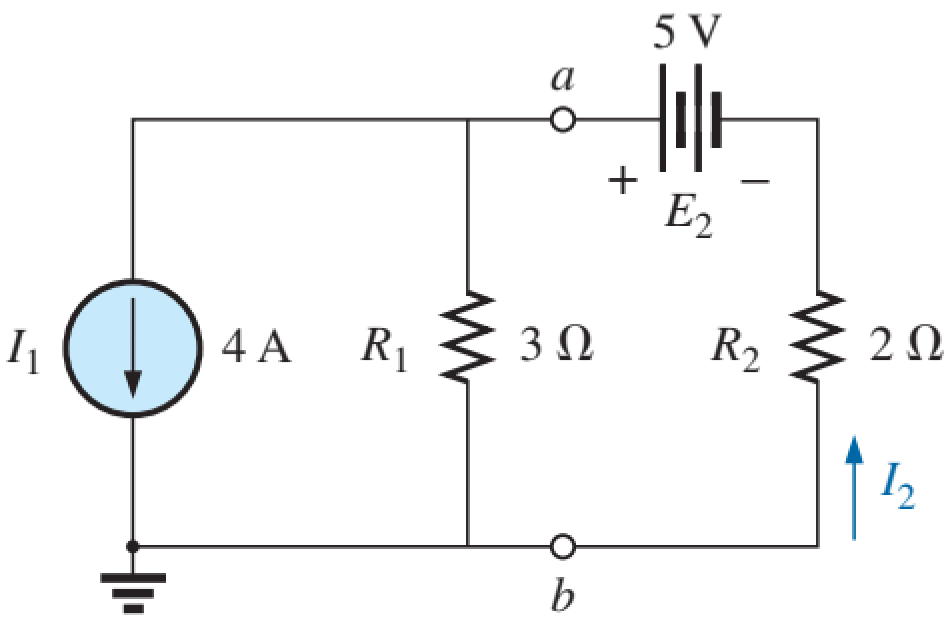


FIG. 9

Two-source network for Example 5.

4. Current Sources in Parallel (1/6)

- ***Two or more current sources in parallel can be replaced by a single current source having a magnitude determined by the difference of the sum of the currents in one direction and the sum in the opposite direction. The new parallel internal resistance is the total resistance of the resulting parallel resistive elements.***
- Consider example 6 next slide.

4. Current Sources in Parallel (2/6)

EXAMPLE 6 Reduce the parallel current sources in Fig. 11 to a single current source.

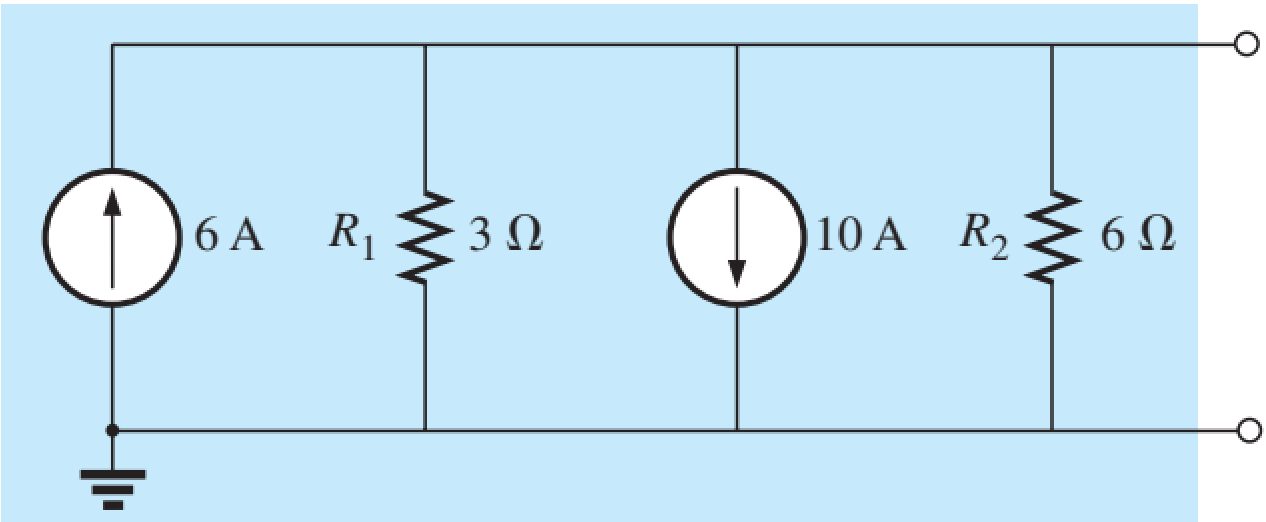


FIG. 11
Parallel current sources for Example 6.

4. Current Sources in Parallel (3/6)

Solution: The net source current is

$$I = 10\text{ A} - 6\text{ A} = \mathbf{4\text{ A}}$$

with the direction being that of the larger source.

The net internal resistance is the parallel combination of resistances, R_1 and R_2:

$$R_p = 3\ \Omega \,\|\, 6\ \Omega = \mathbf{2\ \Omega}$$

The reduced equivalent appears in Fig. 12.

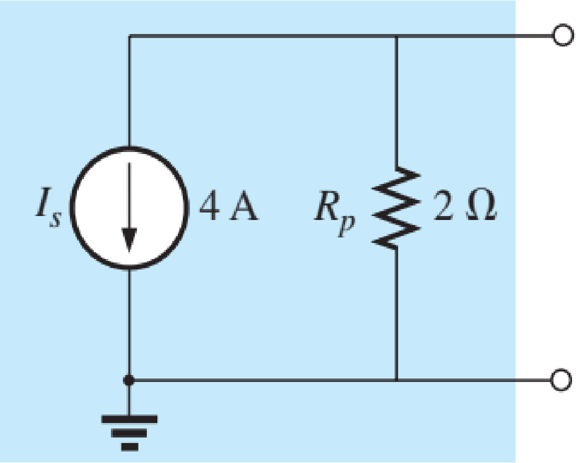


FIG. 12
Reduced equivalent for the configuration of Fig. 11.

4. Current Sources in Parallel (4/6)

EXAMPLE 7 Reduce the parallel current sources in Fig. 13 to a single current source.

Solution: The net current is

$$I = 7\text{ A} + 4\text{ A} - 3\text{ A} = \mathbf{8\text{ A}}$$

with the direction shown in Fig. 14. The net internal resistance remains the same.

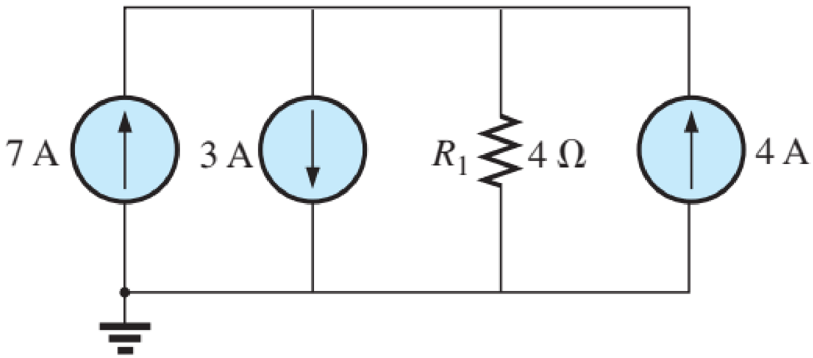


FIG. 13
Parallel current sources for Example 7.

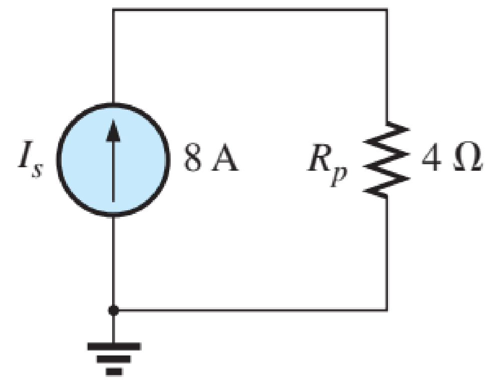


FIG. 14
Reduced equivalent for Fig. 13.

4. Current Sources in Parallel (5/6)

EXAMPLE 8 Reduce the network in Fig. 15 to a single current source, and calculate the current through R_L.

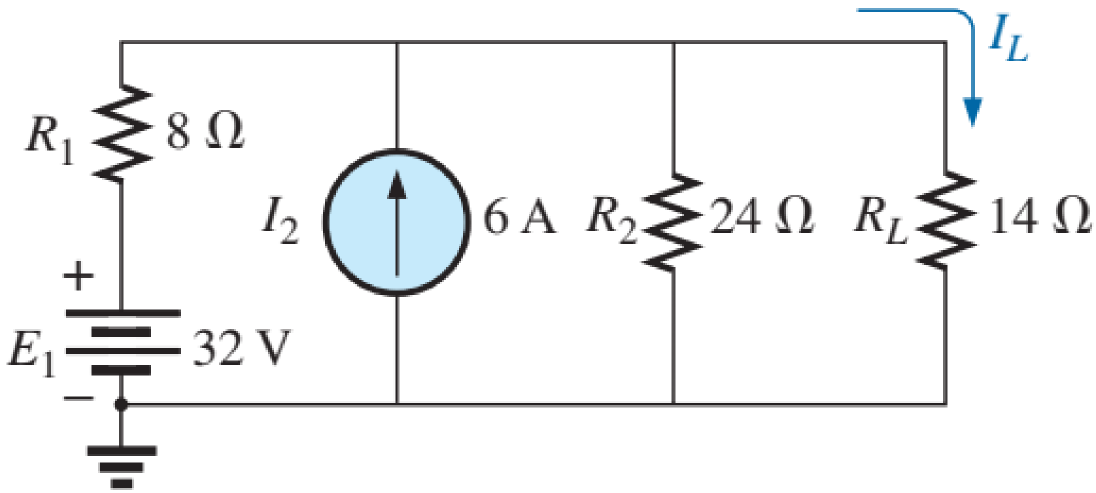


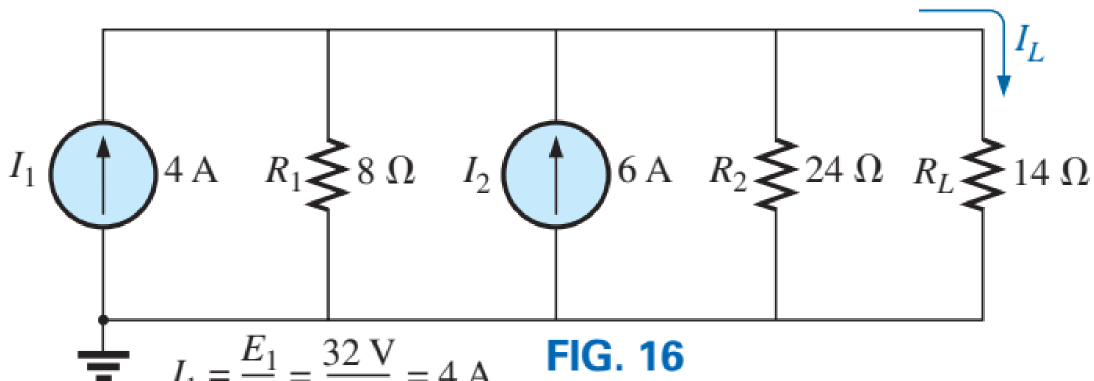
FIG. 15
Example 8.

4. Current Sources in Parallel (6/6)

Solution: In this example, the voltage source will first be converted to a current source as shown in Fig. 16. Combining current sources gives

$$I_s = I_1 + I_2 = 4\text{ A} + 6\text{ A} = \mathbf{10\text{ A}}$$

and $$R_s = R_1 \| R_2 = 8\ \Omega \| 24\ \Omega = \mathbf{6\ \Omega}$$



$$I_1 = \frac{E_1}{R_1} = \frac{32\text{ V}}{8\ \Omega} = 4\text{ A}$$

FIG. 16

Applying the current divider rule to the resulting network in Fig. 17 gives

$$I_L = \frac{R_p I_s}{R_p + R_L} = \frac{(6\ \Omega)(10\text{ A})}{6\ \Omega + 14\ \Omega} = \frac{60\text{ A}}{20} = \mathbf{3\text{ A}}$$

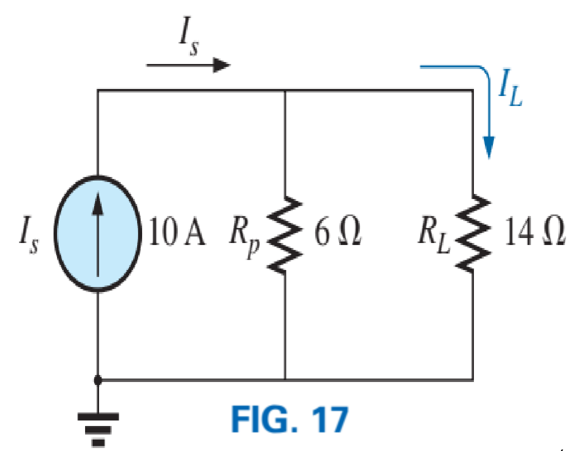


FIG. 17

5. Current Sources in Series

- The current through any branch of a network can be only single-valued. For the situation indicated at point ***a*** in Fig. 18,we find by application of Kirchhoff's current law that the current leaving that point is greater than that entering—an impossible situation. Therefore,
  - ***Current sources of different current ratings are not connected in series.***

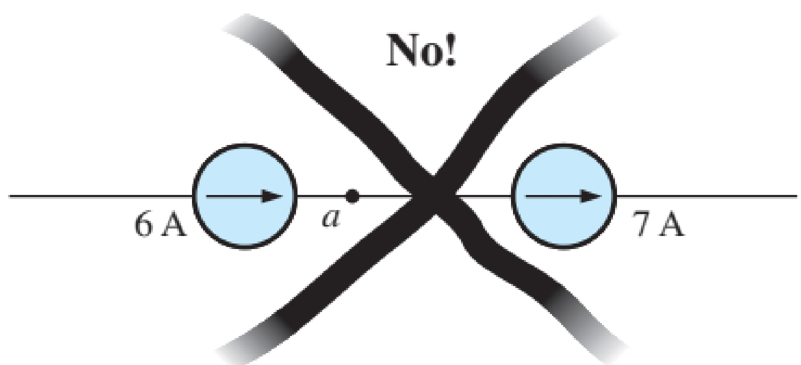


FIG. 18
Invalid situation.

6. Branch-Current Analysis (1/7)

- Branch-current analysis allows us to directly calculate the current in each branch of a circuit. When applying branch-current analysis, you will find the following technique useful.
  1. Arbitrarily assign current directions to each branch in the network. If a particular branch has a current source, then this step is not necessary since you already know the magnitude and direction of the current in this branch.
  2. Using the assigned currents, label the polarities of the voltage drops across all resistors in the circuit.
  3. Apply Kirchhoff's voltage law around each of the closed loops. Write just enough equations to include all branches in the loop equations. If a branch has only a current source and no series resistance, it is not necessary to include it in the KVL equations.
  4. Apply Kirchhoff's current law at enough nodes to ensure that all branch currents have been included. In the event that a branch has only a current source, it will need to be included in this step.
  5. Solve the resulting simultaneous linear equations.

6. Branch-Current Analysis (2/7)

Example 9: Apply the branch-current method to the network in Fig.19.

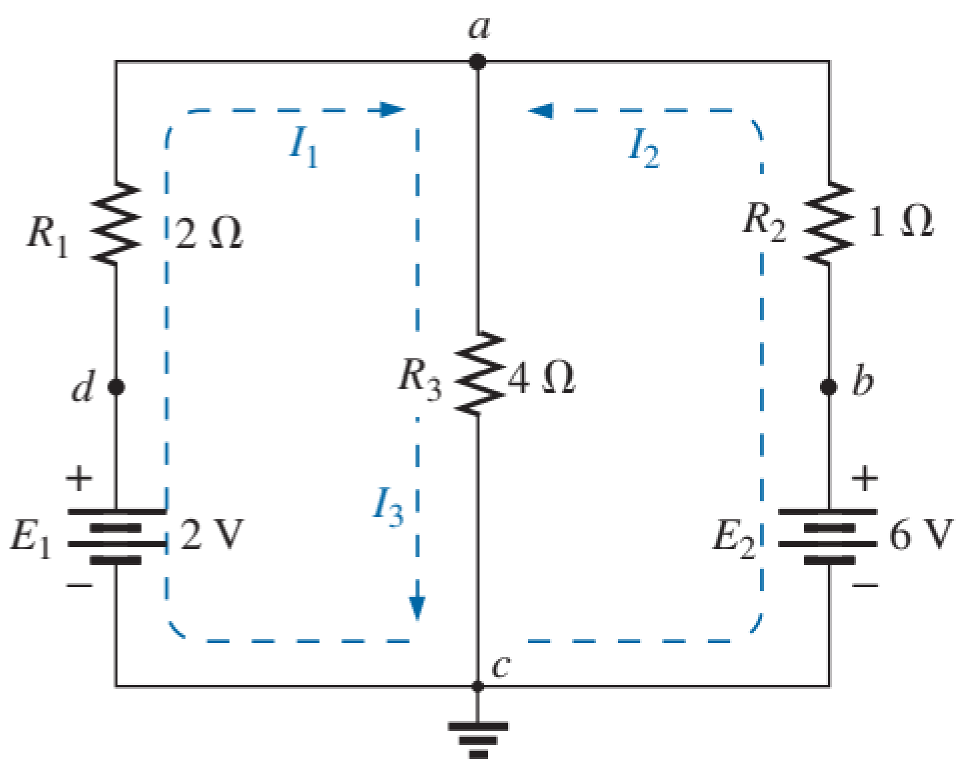


Figure 19

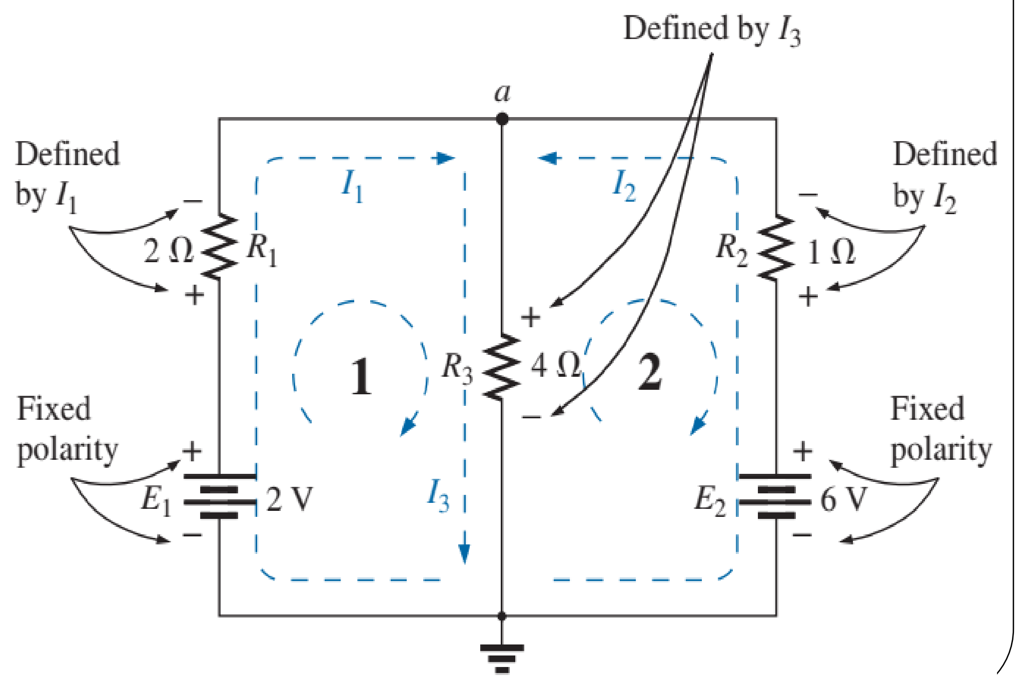
6. Branch-Current Analysis (3/7)

Solution:

Step 1: Since there are three distinct branches (*cda*, *cba*, *ca*), three currents of arbitrary directions (I_1, I_2, I_3) are chosen, as indicated in Fig. ***19***. The current directions for I_1 and I_2 were chosen to match the "pressure" applied by sources E_1 and E_2, respectively. Since both I_1 and I_2 enter node a, I_3 is leaving.

Step 2: Polarities for each resistor are drawn to agree with assumed current directions, as indicated in Fig. ***20***

Figure 20



6. Branch-Current Analysis (4/7)

Step 3: Kirchhoff's voltage law is applied around each closed loop (1 and 2) in the clockwise direction:

Rise in potential

$$\text{loop 1:}\quad \Sigma_{\circlearrowright} V = +E_1 - V_{R_1} - V_{R_3} = 0$$

Drop in potential

and

Rise in potential

$$\text{loop 2:}\quad \Sigma_{\circlearrowright} V = +V_{R_3} + V_{R_2} - E_2 = 0$$

Drop in potential

$$\text{loop 1:}\quad \Sigma_{\circlearrowright} V = \underbrace{+2\ \text{V}}_{\substack{\text{Battery}\\ \text{potential}}} - \underbrace{(2\ \Omega)I_1}_{\substack{\text{Voltage drop}\\ \text{across } 2\ \Omega\\ \text{resistor}}} - \underbrace{(4\ \Omega)I_3}_{\substack{\text{Voltage drop}\\ \text{across } 4\ \Omega\\ \text{resistor}}} = 0$$

$$\text{loop 2:}\quad \Sigma_{\circlearrowright} V = (4\ \Omega)I_3 + (1\ \Omega)I_2 - 6\ \text{V} = 0$$

6. Branch-Current Analysis (5/7)

Step 4: Applying Kirchhoff's current law at node *a* (in a two-node network, the law is applied at only one node) gives

$$I_1 + I_2 = I_3$$

Step 5: There are three equations and three unknowns (units removed for clarity):

$$\begin{aligned} 2 - 2I_1 - 4I_3 &= 0 \\ 4I_3 + 1I_2 - 6 &= 0 \\ I_1 + I_2 &= I_3 \end{aligned} \qquad \text{Rewritten: } \begin{aligned} 2I_1 + 0 + 4I_3 &= 2 \\ 0 + I_2 + 4I_3 &= 6 \\ I_1 + I_2 - I_3 &= 0 \end{aligned}$$

6. Branch-Current Analysis (6/7)

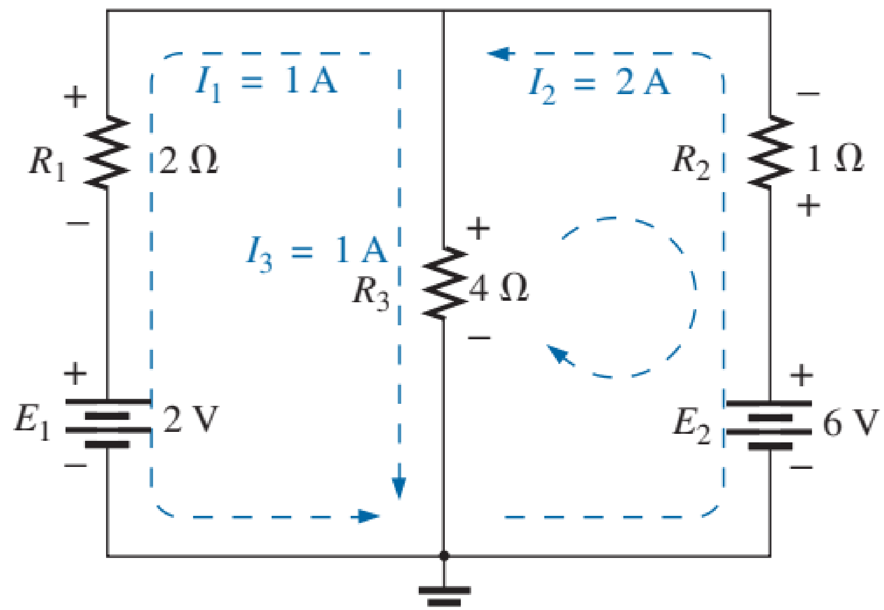
Using third-order determinants, we have

$$I_1 = \frac{\begin{vmatrix} 2 & 0 & 4 \\ 6 & 1 & 4 \\ 0 & 1 & -1 \end{vmatrix}}{D = \begin{vmatrix} 2 & 0 & 4 \\ 0 & 1 & 4 \\ 1 & 1 & -1 \end{vmatrix}} = \mathbf{-1\ A}$$

A negative sign in front of a branch current indicates only that the actual current is in the direction opposite to that assumed.

$$I_2 = \frac{\begin{vmatrix} 2 & 2 & 4 \\ 0 & 6 & 4 \\ 1 & 0 & -1 \end{vmatrix}}{D} = \mathbf{2\ A}$$

$$I_3 = \frac{\begin{vmatrix} 2 & 0 & 2 \\ 0 & 1 & 6 \\ 1 & 1 & 0 \end{vmatrix}}{D} = \mathbf{1\ A}$$



Reviewing the results of the analysis of the network in Fig. 19.

6. Branch-Current Analysis (7/7)

Homework!

Example 10: Apply the branch-current method to the network in Fig.21.

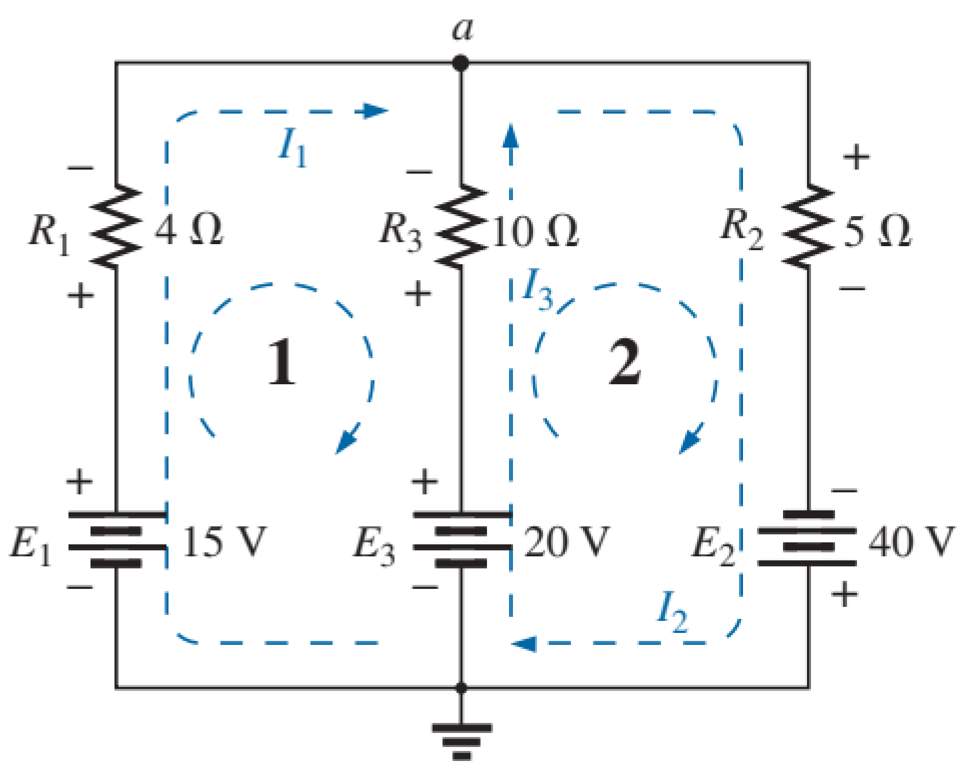


Figure 21

7. Mesh Analysis (1/10)

- A better approach and one that is used extensively in analysing linear bilateral networks is called ***mesh***(or ***loop***) ***analysis***. While the technique is similar to branch-current analysis, the number of simultaneous linear equations tends to be less. The principal difference between mesh analysis and branch-current analysis is that we simply need to apply Kirchhoff's voltage law around closed loops without the need for applying Kirchhoff's current law. The steps used in solving a circuit using mesh analysis are as follows:

1. Arbitrarily assign a *clockwise current* to each interior closed loop in the network. Although the assigned current may be in any direction, a clock-wise direction is used to make later work simpler.
2. Using the assigned loop currents, *indicate the voltage polarities across all resistors in the circuit*. *For a resistor that is common to two loops, the polarities of the voltage drop due to each loop current should be indicated on the appropriate side of the component*.
3. *Applying Kirchhoff's voltage law*, write the loop equations for each loop in the network. *Do not forget that resistors that are common to two loops will have two voltage drops, one due to each loop*.
4. Solve the resultant simultaneous linear equations.
5. Branch currents are determined by algebraically combining the loop currents that are common to the branch.

7. Mesh Analysis (2/10)

Example 11: Find the current through each branch of the network in Fig.22.

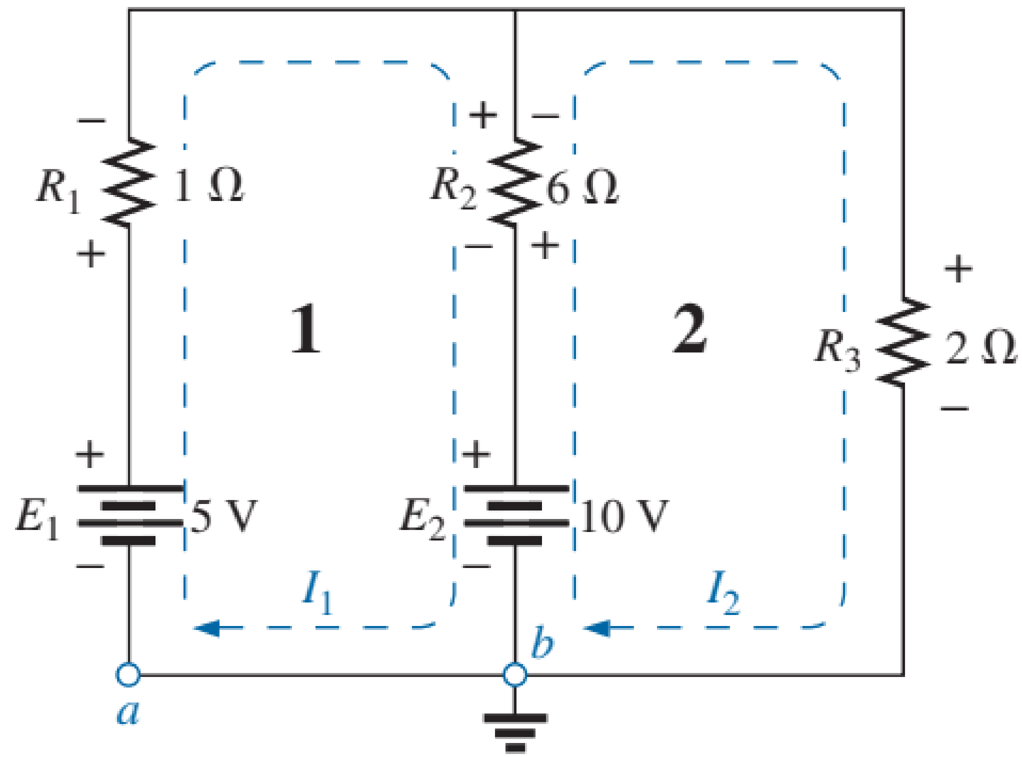


Figure 22

7. Mesh Analysis (3/10)

Solution:

Steps 1 and 2: These are as indicated in the circuit. Note that the polarities of the 6 Ω resistor are different for each loop current.

Step 3: Kirchhoff's voltage law is applied around each closed loop in the clockwise direction:

loop 1: $+E_1 - V_1 - V_2 - E_2 = 0$ (clockwise starting at point a)

$$+5\text{ V} - (1\ \Omega)I_1 - (6\ \Omega)(I_1 \underset{\uparrow}{-} I_2) - 10\text{ V} = 0$$

I_2 flows through the 6 Ω resistor in the direction opposite to I_1.

loop 2: $E_2 - V_2 - V_3 = 0$ (clockwise starting at point b)

$$+10\text{ V} - (6\ \Omega)(I_2 - I_1) - (2\ \Omega)I_2 = 0$$

The equations are rewritten as

$$\left.\begin{aligned} 5 - I_1 - 6I_1 + 6I_2 - 10 &= 0 \\ 10 - 6I_2 + 6I_1 - 2I_2 &= 0 \end{aligned}\right\} \begin{aligned} -7I_1 + 6I_2 &= 5 \\ +6I_1 - 8I_2 &= -10 \end{aligned}$$

7. Mesh Analysis (4/10)

Step 4:

$$I_1 = \frac{\begin{vmatrix} 5 & 6 \\ -10 & -8 \end{vmatrix}}{\begin{vmatrix} -7 & 6 \\ 6 & -8 \end{vmatrix}} = \frac{-40 + 60}{56 - 36} = \frac{20}{20} = \mathbf{1\ A}$$

$$I_2 = \frac{\begin{vmatrix} -7 & 5 \\ 6 & -10 \end{vmatrix}}{20} = \frac{70 - 30}{20} = \frac{40}{20} = \mathbf{2\ A}$$

Since I_1 and I_2 are positive and flow in opposite directions through the $6\ \Omega$ resistor and $10\ \text{V}$ source, the two currents in the direction of the larger:

$$I_2 > I_1 \quad (2\ \text{A} > 1\ \text{A})$$

Therefore,

$$I_{R_2} = I_2 - I_1 = 2\ \text{A} - 1\ \text{A} = \mathbf{1\ A} \quad \text{in the direction of } I_2$$

7. Mesh Analysis (5/10)

Homework!

Example 12: Find the branch currents of the networks in Fig.23.

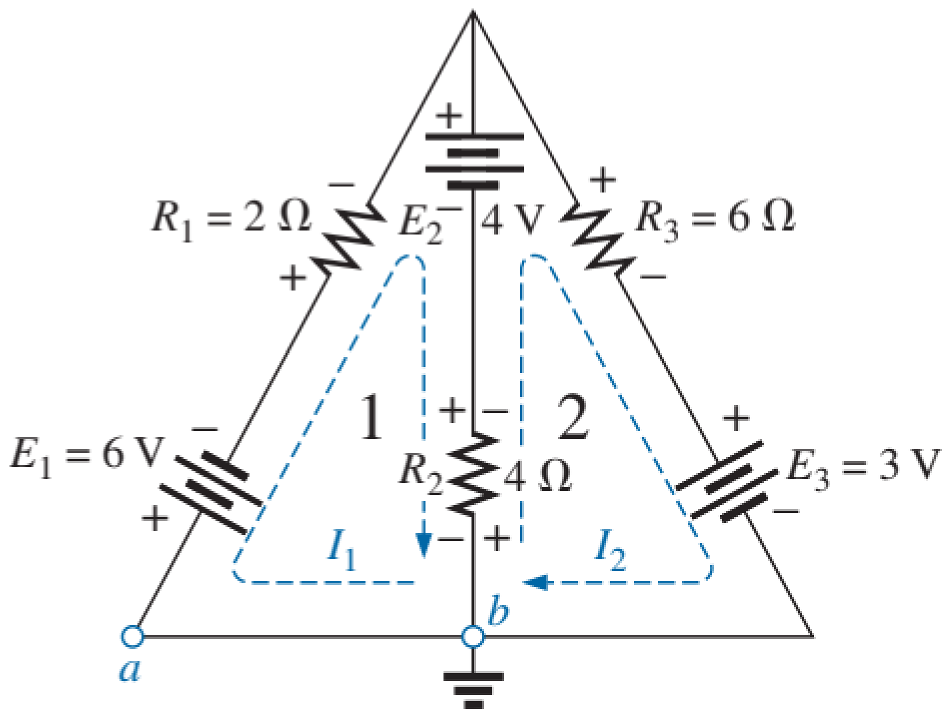


Figure 23

7. Mesh Analysis (6/10)

- ***Supermesh Currents-*** Occasionally, you will find current sources in a network without a parallel resistance. This removes the possibility of converting the source to a voltage source as required by the given procedure. In such cases, you have a choice of using ***supermesh approach***.
- Start as before, and assign a mesh current to each independent loop, including the current sources, as if they were resistors or voltage sources. Then mentally (redraw the network if necessary) remove the current sources (replace with open-circuit equivalents),and apply Kirchhoff's voltage law to all the remaining independent paths of the network using the mesh currents just defined. Any resulting path, including two or more mesh currents, is said to be the path of a supermesh current. Then relate the chosen mesh currents of the network to the independent current sources of the network, and solve for the mesh currents.

7. Mesh Analysis (7/10)

Example 13: Using mesh analysis, determine the currents of the network in Fig.24.

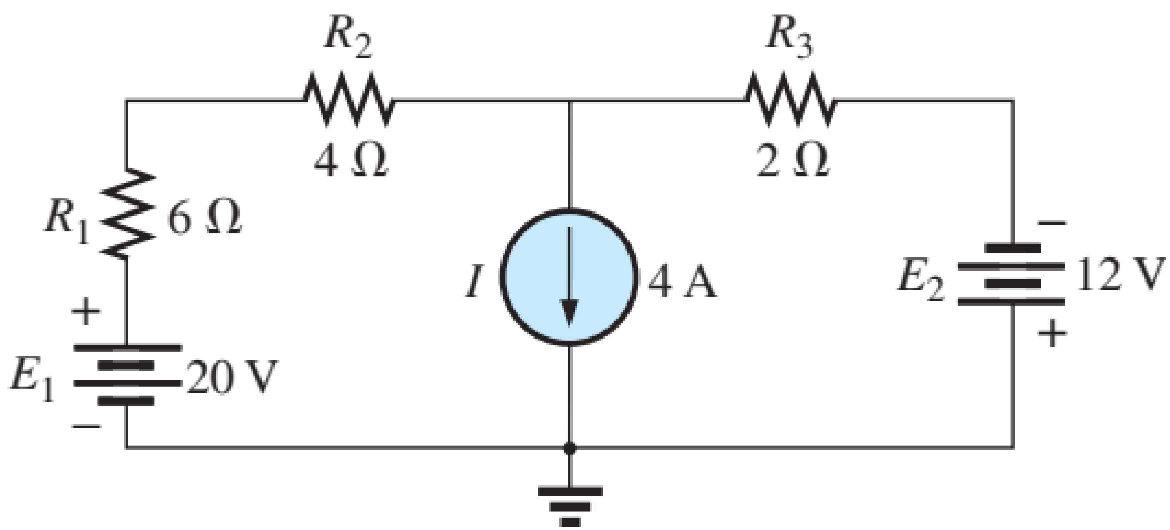


Figure 24

7. Mesh Analysis (8/10)

Solution: First, the mesh currents for the network are defined, as shown in Fig. **25** Then the current source is mentally removed, as shown in Fig **26** , and Kirchhoff's voltage law is applied to the resulting network. The single path now including the effects of two mesh currents is referred to as the path of a *supermesh current*.

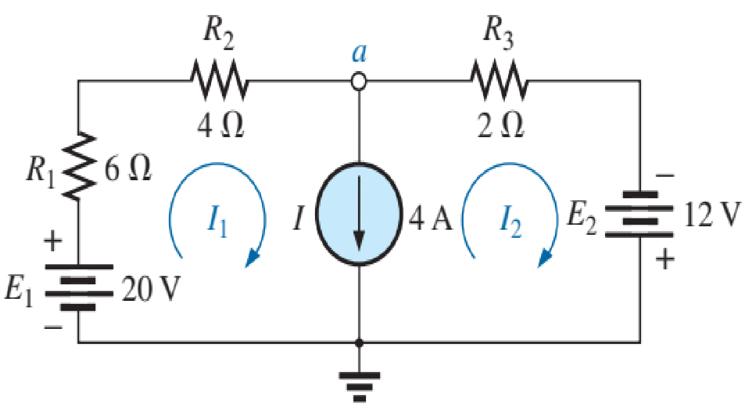


Figure 25

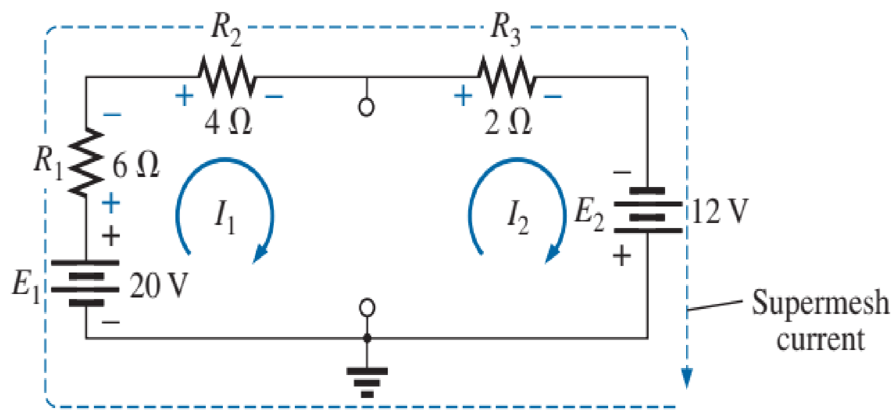


Figure 26

7. Mesh Analysis (9/10)

Applying Kirchhoff's law gives

$$20\text{ V} - I_1(6\,\Omega) - I_1(4\,\Omega) - I_2(2\,\Omega) + 12\text{ V} = 0$$

or

$$10I_1 + 2I_2 = 32$$

Node a is then used to relate the mesh currents and the current source using Kirchhoff's current law:

$$I_1 = I + I_2$$

The result is two equations and two unknowns:

$$\begin{aligned} 10I_1 + 2I_2 &= 32 \\ I_1 - I_2 &= 4 \end{aligned}$$

Applying determinants gives

$$I_1 = \frac{\begin{vmatrix} 32 & 2 \\ 4 & -1 \end{vmatrix}}{\begin{vmatrix} 10 & 2 \\ 1 & -1 \end{vmatrix}} = \frac{(32)(-1) - (2)(4)}{(10)(-1) - (2)(1)} = \frac{40}{12} = \mathbf{3.33\ A}$$

and

$$I_2 = I_1 - I = 3.33\text{ A} - 4\text{ A} = \mathbf{-0.67\ A}$$

7. Mesh Analysis (10/10)

Homework!

Example 14: Using mesh analysis, determine the currents for the network in Fig. 27.

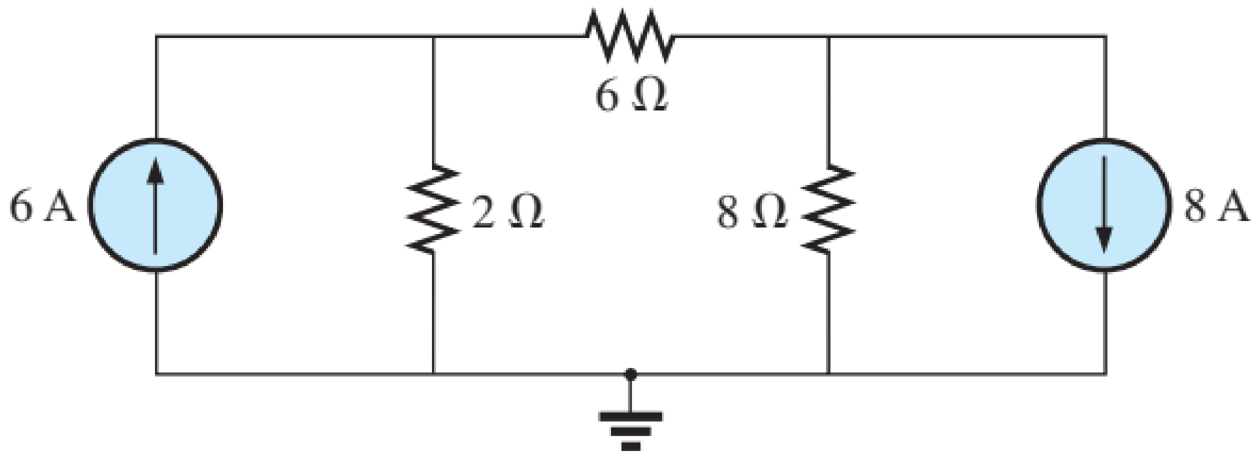


Figure 27

8. Nodal Analysis (1/7)

- The methods introduced previously thus far have all been to find the currents of the network. We now turn our attention to ***nodal analysis*** -a method that provides the nodal voltages of a network, that is, the voltage from the various ***nodes*** (junction points) of the network to ground. The method is developed through the use of Kirchhoff's current law.

8. Nodal Analysis (2/7)

Nodal Analysis Procedure

1. Determine the number of nodes within the network.
2. Pick a reference node, and label each remaining node with a sub-scripted value of voltage: V_1, V_2, and so on.
3. Apply Kirchhoff's current law at each node except the reference. Assume that all unknown currents leave the node for each application of Kirchhoff's current law.
4. Solve the resulting equations for the nodal voltages.

8. Nodal Analysis (3/7)

Example 15: Apply nodal analysis to the network in Fig.28.

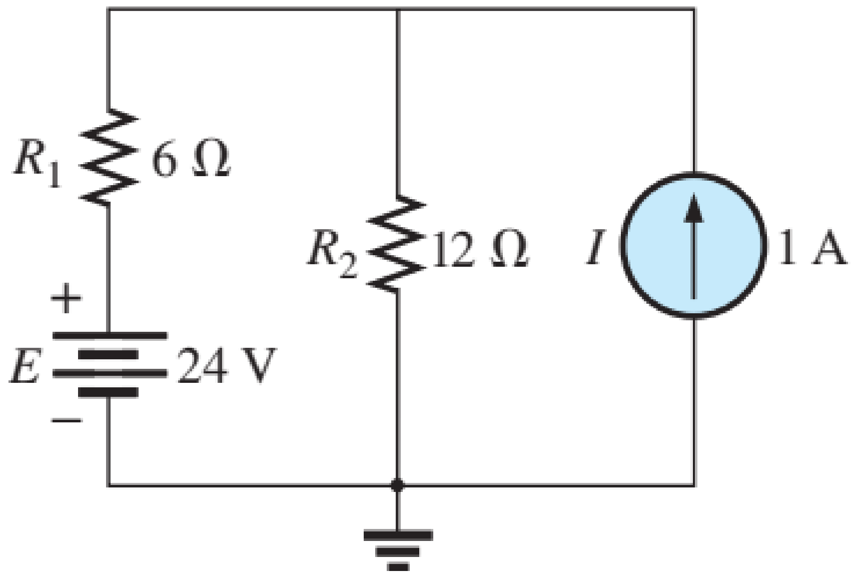


Figure 28

8. Nodal Analysis (4/7)

Solution:

Steps 1 and 2: The network has two nodes, as shown in Fig.**29** The lower node is defined as the reference node at ground potential (zero volts), and the other node as V_1, the voltage from node 1 to ground.

Step 3: I_1 and I_2 are defined as leaving the node in Fig. **30**, and Kirchhoff's current law is applied as follows:

$$I = I_2 + I_1$$

The current I_2 is related to the nodal voltage V_1 by Ohm's law:

$$I_2 = \frac{V_{R_2}}{R_2} = \frac{V_1}{R_2}$$

The current I_1 is also determined by Ohm's law as follows:

$$I_1 = \frac{V_{R_1}}{R_1}$$

with $$V_{R_1} = V_1 - E$$

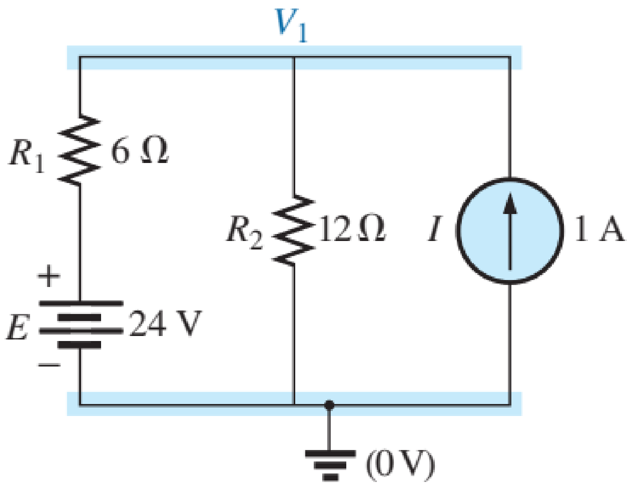


Figure 29

8. Nodal Analysis (5/7)

Substituting into the Kirchhoff's current law equation

$$I = \frac{V_1 - E}{R_1} + \frac{V_1}{R_2}$$

and rearranging, we have

$$I = \frac{V_1}{R_1} - \frac{E}{R_1} + \frac{V_1}{R_2} = V_1\left(\frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{E}{R_1}$$

or

$$V_1\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{E}{R_1} + I$$

Substituting numerical values, we obtain

$$V_1\left(\frac{1}{6\,\Omega} + \frac{1}{12\,\Omega}\right) = \frac{24\text{ V}}{6\,\Omega} + 1\text{ A} = 4\text{ A} + 1\text{ A}$$

$$V_1\left(\frac{1}{4\,\Omega}\right) = 5\text{ A}$$

$$V_1 = \mathbf{20\ V}$$

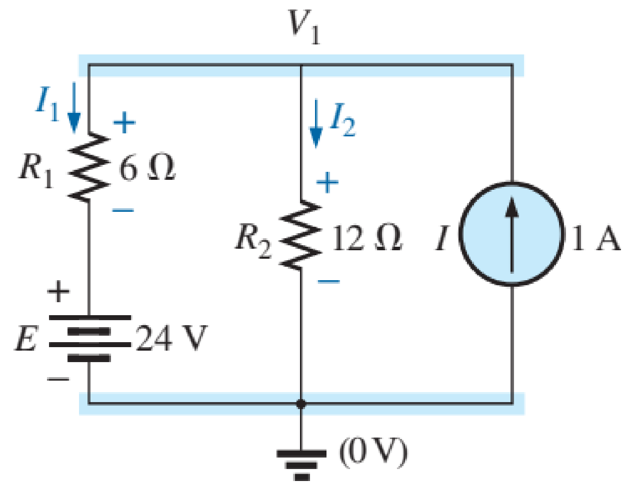


Figure 30

8. Nodal Analysis (6/7)

The currents I_1 and I_2 can then be determined by using the preceding equations:

$$I_1 = \frac{V_1 - E}{R_1} = \frac{20\text{ V} - 24\text{ V}}{6\ \Omega} = \frac{-4\text{ V}}{6\ \Omega}$$

$$= \mathbf{-0.67\ A}$$

The minus sign indicates that the current I_1 has a direction opposite to that appearing in Fig. **30** . In addition,

$$I_2 = \frac{V_1}{R_2} = \frac{20\text{ V}}{12\ \Omega} = \mathbf{1.67\ A}$$

8. Nodal Analysis (7/7)

Homework!

Example 16: Apply nodal analysis to the networks in Fig. 31 and Fig. 32.

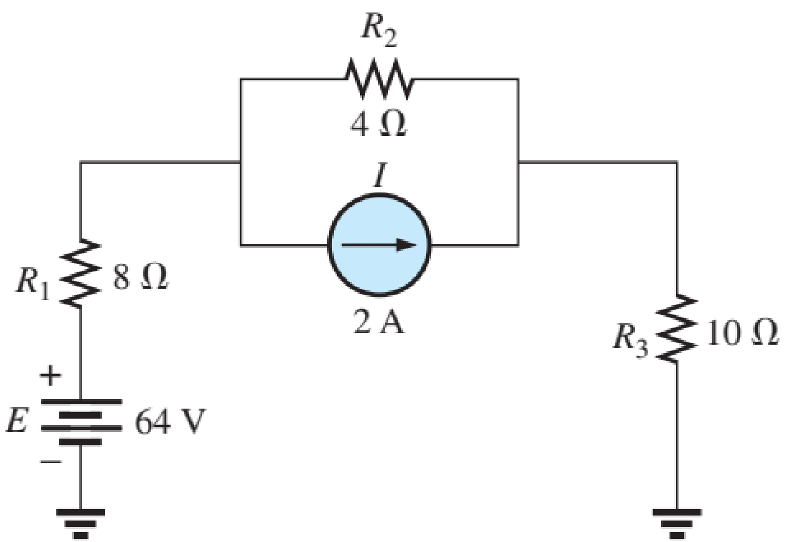


Figure 31

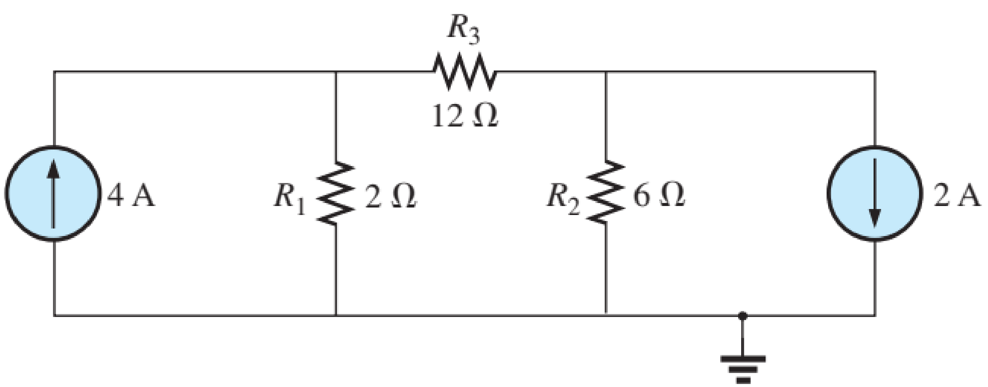


Figure 32

9. Bridge Networks (1/3)

- This section introduces the bridge network, a configuration that has a multitude of applications. This type of network is used in both dc and ac meters. Electronics courses introduce these in the discussion of rectifying circuits used in converting a varying signal to one of a steady nature (such as dc).
- The bridge network may appear in one of the three forms as indicated in Fig. 32(c). The network in Fig. 32(c) is also called a *symmetrical lattice network* if $R_2 = R_3$ and $R_1 = R_4$.

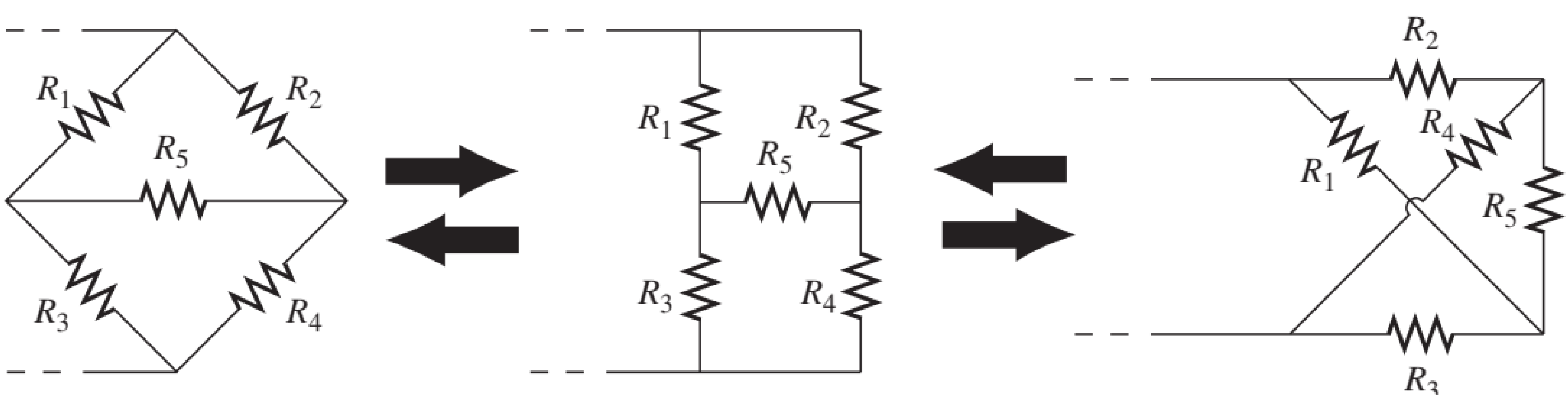
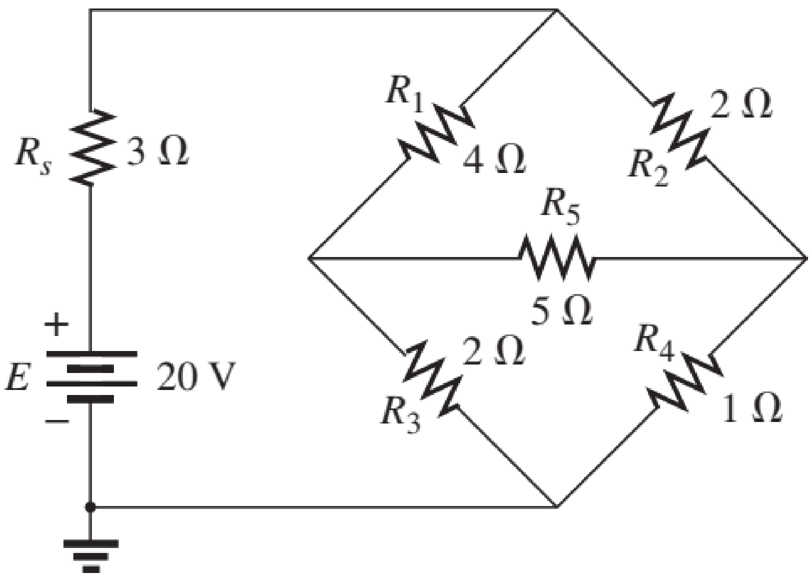


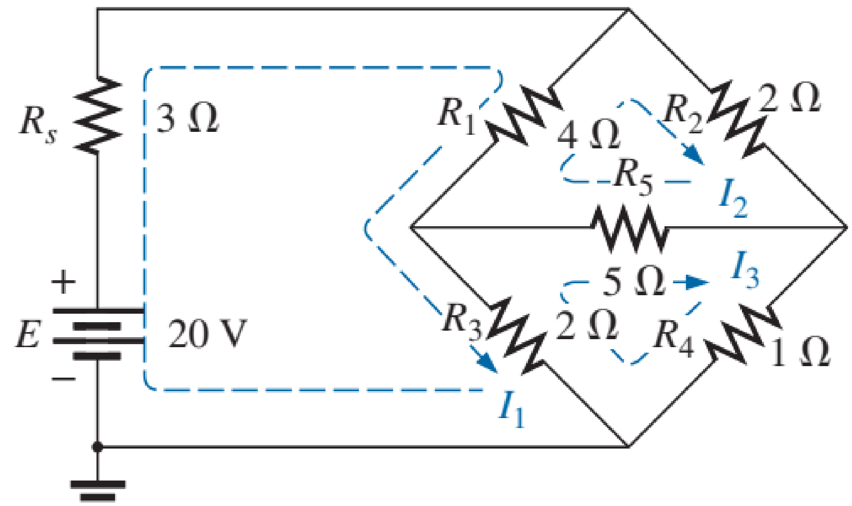
Figure 32 (a) ***Figure 32*** (b) ***Figure 32*** (c)

9. Bridge Networks (2/3)

- Let us examine the network in Fig given below using mesh analysis.



Standard bridge configuration.



Assigning the mesh currents to the network

9. Bridge Networks (3/3)

- Mesh Analysis yields:

$$(3\,\Omega + 4\,\Omega + 2\,\Omega)I_1 - (4\,\Omega)I_2 - (2\,\Omega)I_3 = 20\text{ V}$$
$$(4\,\Omega + 5\,\Omega + 2\,\Omega)I_2 - (4\,\Omega)I_1 - (5\,\Omega)I_3 = 0$$
$$(2\,\Omega + 5\,\Omega + 1\,\Omega)I_3 - (2\,\Omega)I_1 - (5\,\Omega)I_2 = 0$$

and

$$9I_1 - 4I_2 - 2I_3 = 20$$
$$-4I_1 + 11I_2 - 5I_3 = 0$$
$$-2I_1 - 5I_2 + 8I_3 = 0$$

with the result that

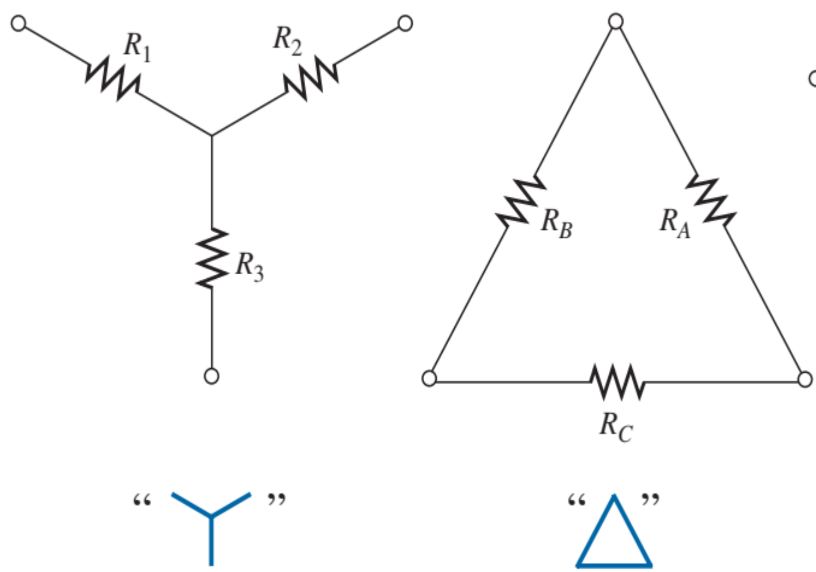
$$I_1 = \mathbf{4\ A}$$
$$I_2 = \mathbf{2.67\ A}$$
$$I_3 = \mathbf{2.67\ A}$$

The net current through the 5 Ω resistor is

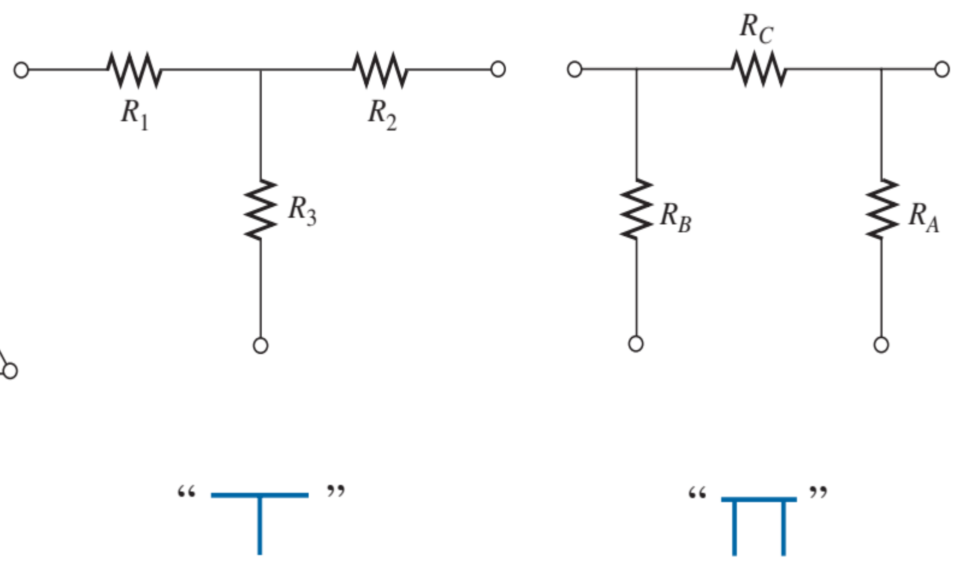
$$I_{5\Omega} = I_2 - I_3 = 2.67\text{ A} - 2.67\text{ A} = 0\text{ A}$$

10. Y-Δ and Δ-Y Conversions (1/9)

- Circuit configurations are often encountered in which the resistors do not appear to be in series or parallel. Under these conditions, it may be necessary to convert the circuit from one form to another to solve for any unknown quantities if mesh or nodal analysis is not applied. Two circuit configurations that often account for these difficulties are the wye (Y) and delta (Δ) configurations depicted in figure given below:



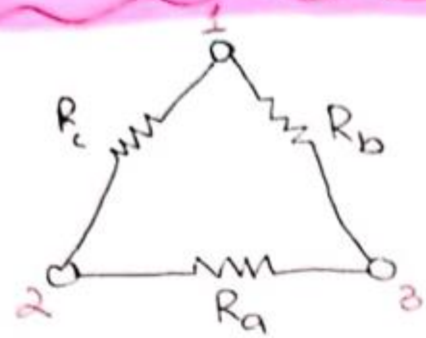
(a)

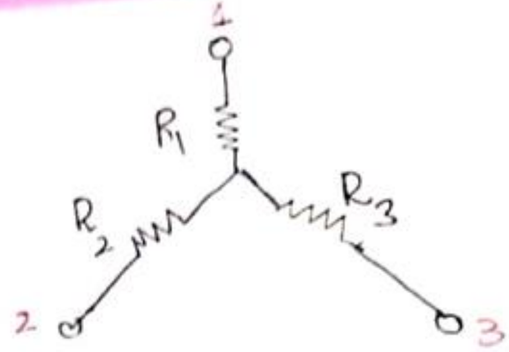


(b)

10. Y-Δ and Δ-Y Conversions (2/9)

Delta (Δ) to Wye (Y) Conversion





$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

10. Y-Δ and Δ-Y Conversions (3/9)

Example 17

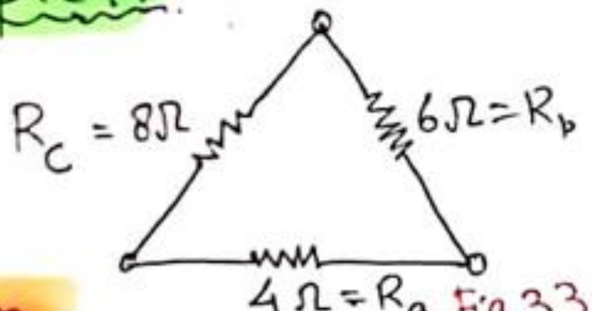
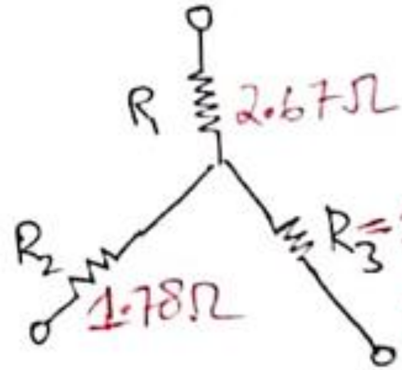


Fig 33



$R_1 = ?$
$R_2 = ?$
$R_3 = ?$

Fig 34

Solution

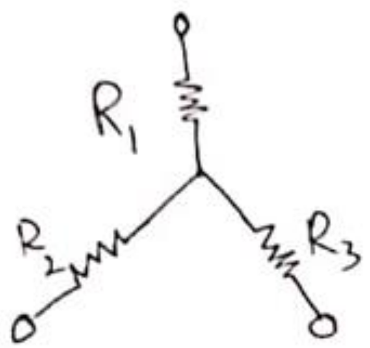
$$\rightarrow R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{8\Omega \times 6\Omega}{8\Omega + 6\Omega + 4\Omega} = 2.67\Omega$$

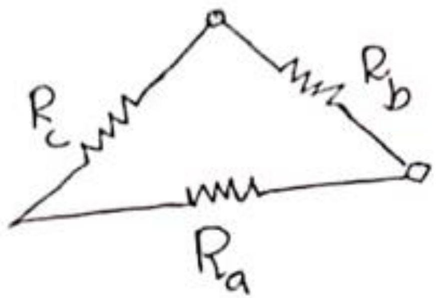
$$\rightarrow R_2 = \frac{R_a R_c}{R_a + R_b + R_c} = \frac{8\Omega \times 4\Omega}{8\Omega + 4\Omega + 6\Omega} = 1.78\Omega$$

$$\rightarrow R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{4\Omega \times 6\Omega}{8\Omega + 4\Omega + 6\Omega} = \frac{24\Omega}{18\Omega} = 1.33\Omega$$

10. Y-Δ and Δ-Y Conversions (4/9)

Wye (Y) to Delta (Δ) Conversion.





$$R_a = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

$$R_b = R_1 + R_3 + \frac{R_1 R_3}{R_2}$$

$$R_c = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

10. Y-Δ and Δ-Y Conversions (5/9)

Example 18

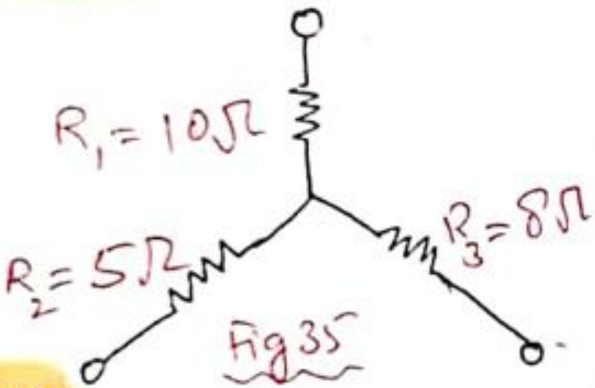


Fig 35

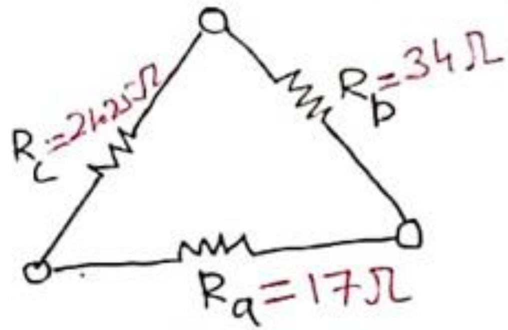


Fig 36

$R_a = ?$
$R_b = ?$
$R_c = ?$

Solution:

$$R_a = R_2 + R_3 + \frac{R_2 R_3}{R_1} = 5\Omega + 8\Omega + \frac{(5\Omega)(8\Omega)}{10\Omega} = 17\Omega$$

$$R_b = R_1 + R_3 + \frac{R_1 R_3}{R_2} = 10\Omega + 8\Omega + \frac{(10\Omega)(8\Omega)}{5\Omega} = 18 + 16 = 34\Omega$$

$$R_c = R_1 + R_2 + \frac{R_1 R_2}{R_3} = 10\Omega + 5\Omega + \frac{(10\Omega)(5\Omega)}{8} = 21.25\Omega$$

10. Y-Δ and Δ-Y Conversions (6/9)

Example 19 Convert Δ in figure below to a Y.

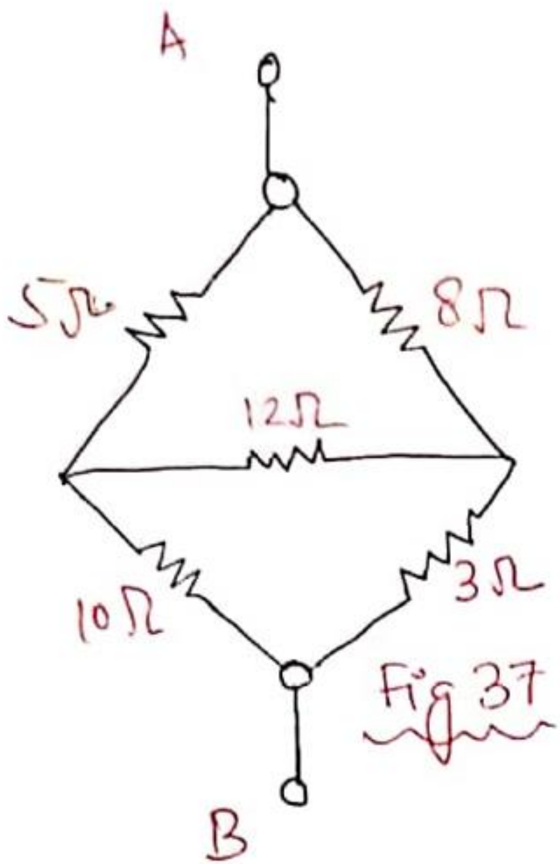


Fig 37

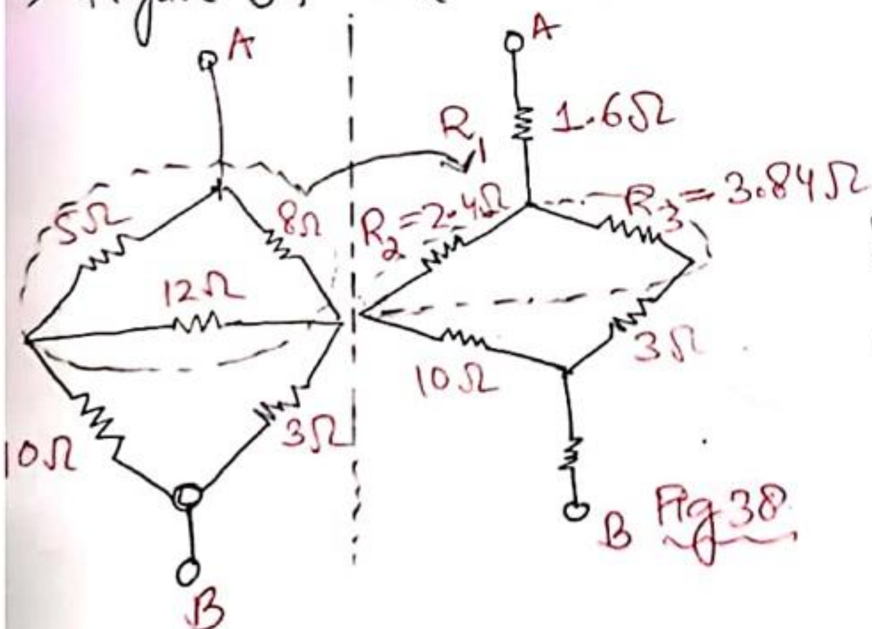
10. Y-Δ and Δ-Y Conversions (7/9)

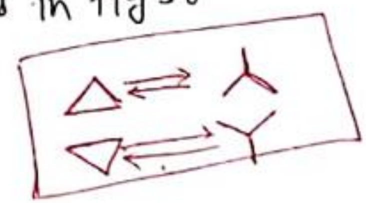
Solution

* 1st, we are supposed to convert Δ into Y
* 2nd, Network needs to convert it into a single resistance Network (R_T)

→ The given network in Fig. 37 seems to be neither in series form nor in parallel form.

→ Figure 37 can also be drawn as in Fig 38





$$R_1 = \frac{5\Omega \times 8\Omega}{5\Omega + 8\Omega + 12\Omega} = \frac{40\Omega}{25\Omega} = 1.6\,\Omega$$

$$R_2 = \frac{5\Omega \times 12\Omega}{5\Omega + 8\Omega + 12\Omega} = \frac{60\Omega}{25\Omega} = 2.4\,\Omega$$

$$R_3 = \frac{12\Omega \times 8\Omega}{5\Omega + 8\Omega + 12\Omega} = \frac{96\Omega}{25\Omega} = 3.84\,\Omega$$

10. Y-Δ and Δ-Y Conversions (8/9)

Network in Figure 38 can be redrawn as: (4)

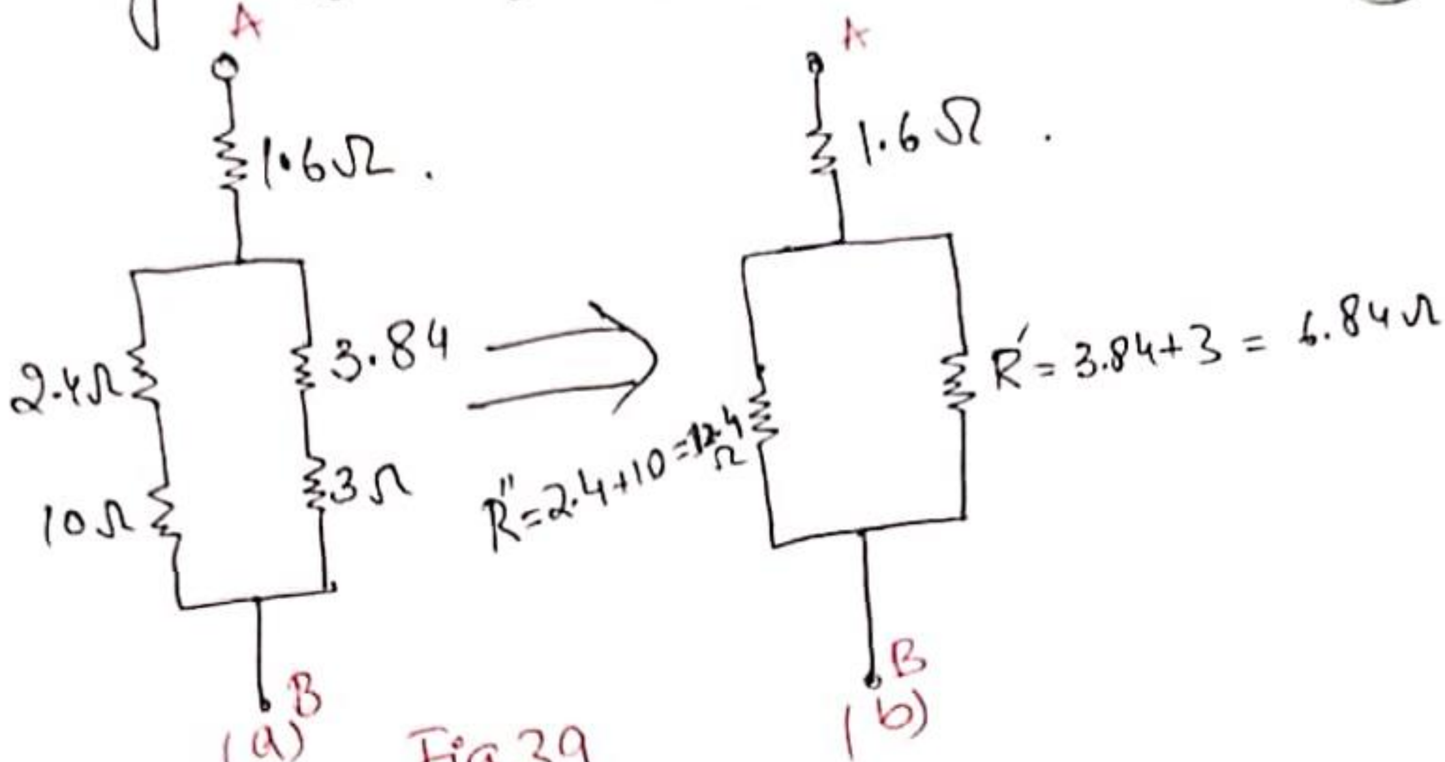


Fig 39

10. Y-Δ and Δ-Y Conversions (9/9)

Now again reduction of n/ω in Fig 39 is done

$$R''' = R'' \| R' = \frac{R'' \cdot R'}{R''+R'} = \frac{(12.4)(6.84)}{(12.4+6.84)} = 4.408\Omega$$

So $R_T = 1.6\Omega + R''' = 1.6\Omega + 4.408\Omega = 6.008\Omega$

The final reduced form of network given in Fig 37 is:

$R_T = 6.008\,\Omega$.

Fig 40

References: Textbooks

1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12th Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5th Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*